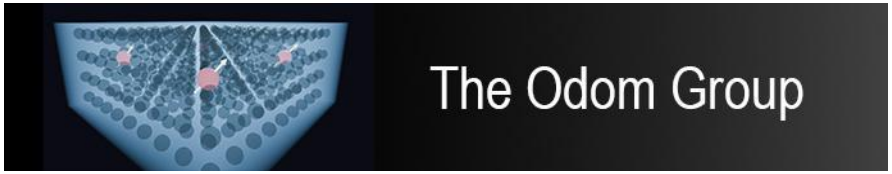
A Dual-Species Beam Source for Doped Cryogenic Crystal Growth

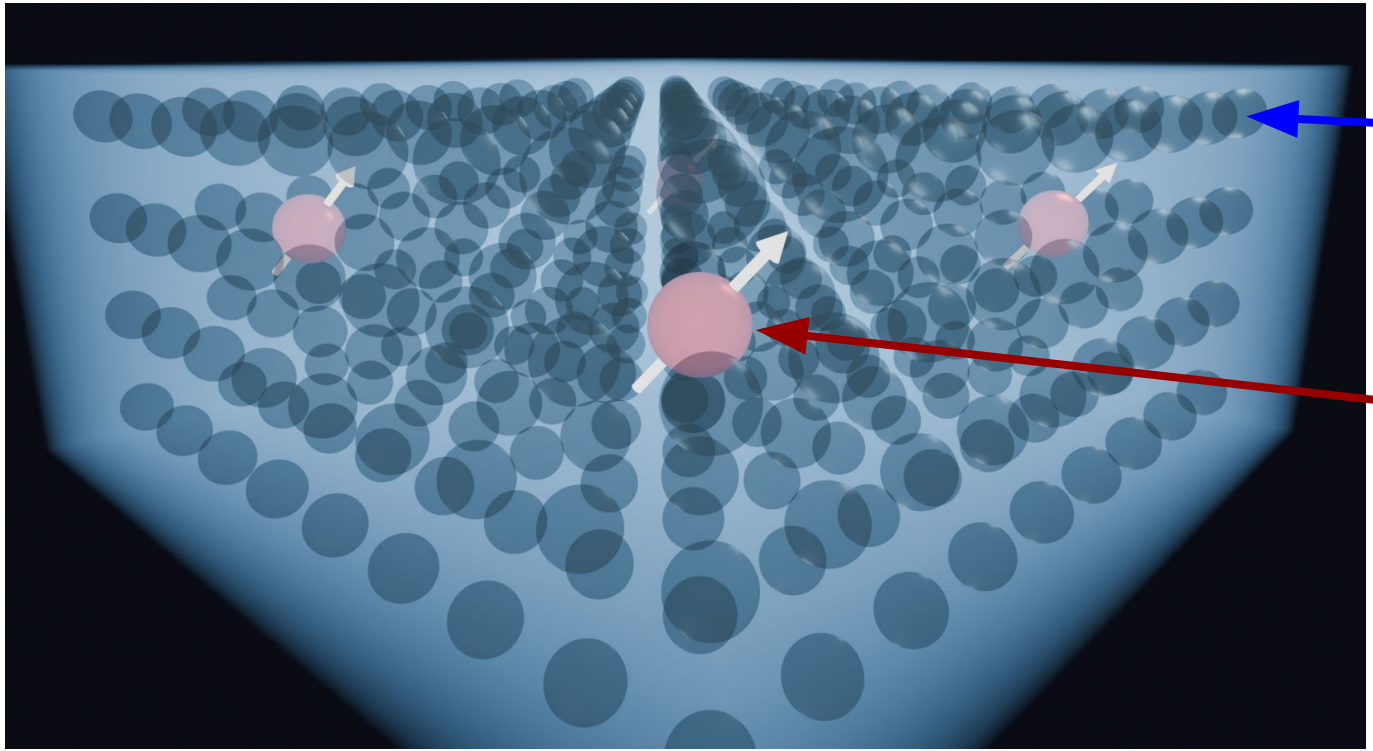
Isaac K. Cross

04/14/2026 CFP Pre-talk



Argonne NATIONAL LABORATORY

Doped Cryogenic Crystal



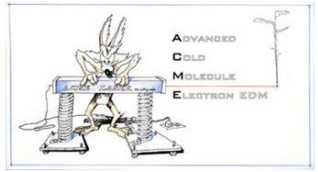
Inert Cryogenic Noble Gas Crystal Host (e.g. Argon)

Dopant to be probed (e.g. Rb),

*Not to scale

Leading EDM Platforms

Beams:



T2* = 2-3 ms

N = 10^{12}

ACME Collaboration
Nature **562**, 355-360 (2018)

Traps:

JILA

T2* = 5 s

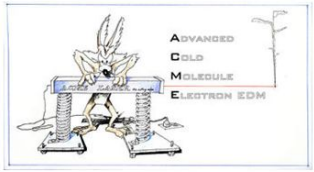
N = 10^{7}

JILA. Eric Cornell
slides ITAMP 2024

General EDM sensitivity: $\sigma \propto \frac{1}{T_2^* \sqrt{N}}$

Leading EDM Platforms

Beams:



$T2^* = 2\text{-}3$ ms

$N = 10^{12}$

ACME Collaboration
Nature **562**, 355-360 (2018)

Traps:

JILA

$T2^* = 5$ s

$N = 10^7$

JILA. Eric Cornell
slides ITAMP 2024

Doped Crystals?

Rb in Ne:

$T2 = 0.1$ s

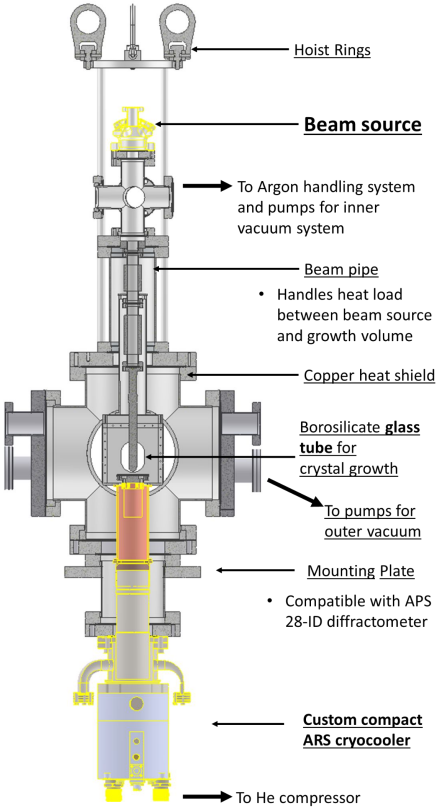
$T2^* = 10$ μs

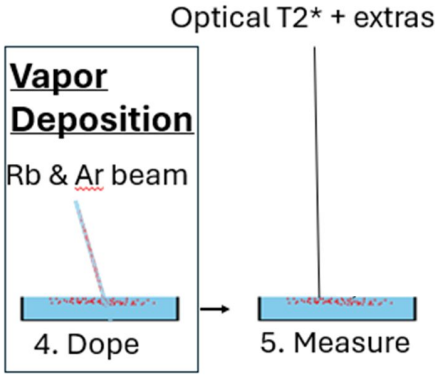
10^{16} atoms/cm^3

J.D. Weinstein et al.,
Phys. Rev. A **104**, 032611 (2021)

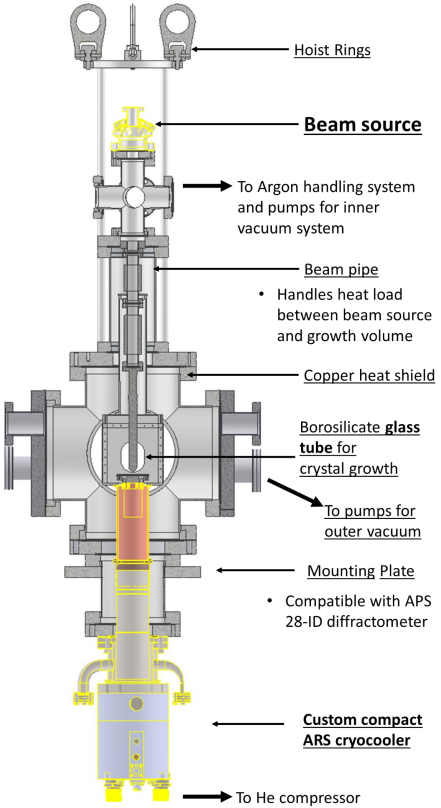
General EDM sensitivity: $\sigma \propto \frac{1}{T_2^* \sqrt{N}}$

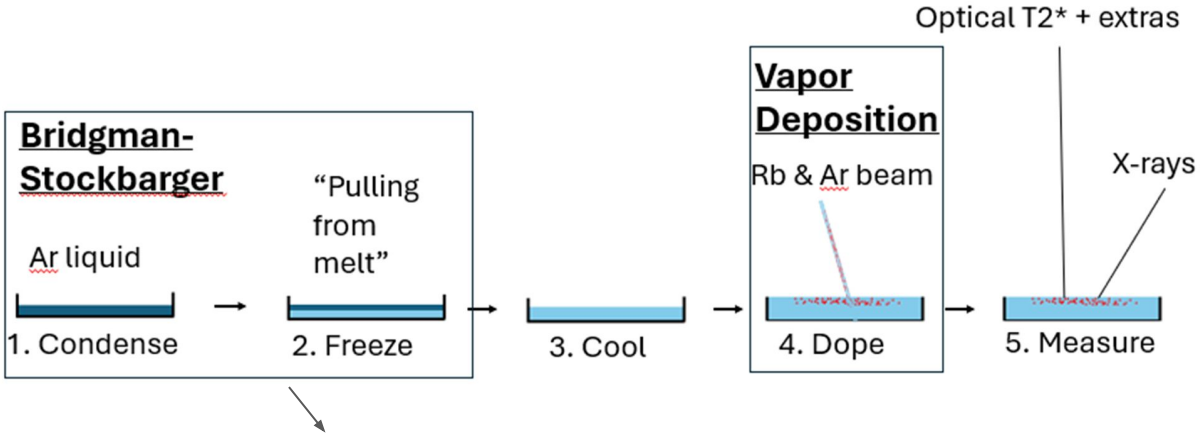
Our Growth Method





Our Growth Method

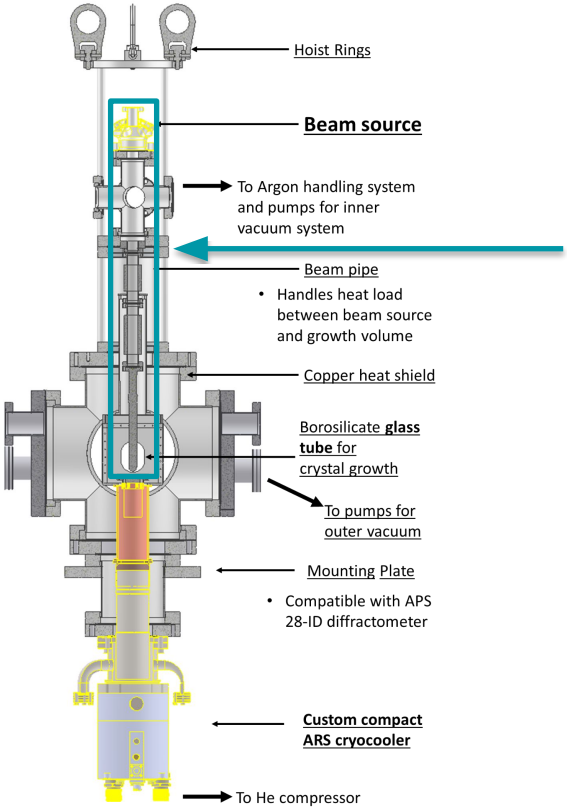


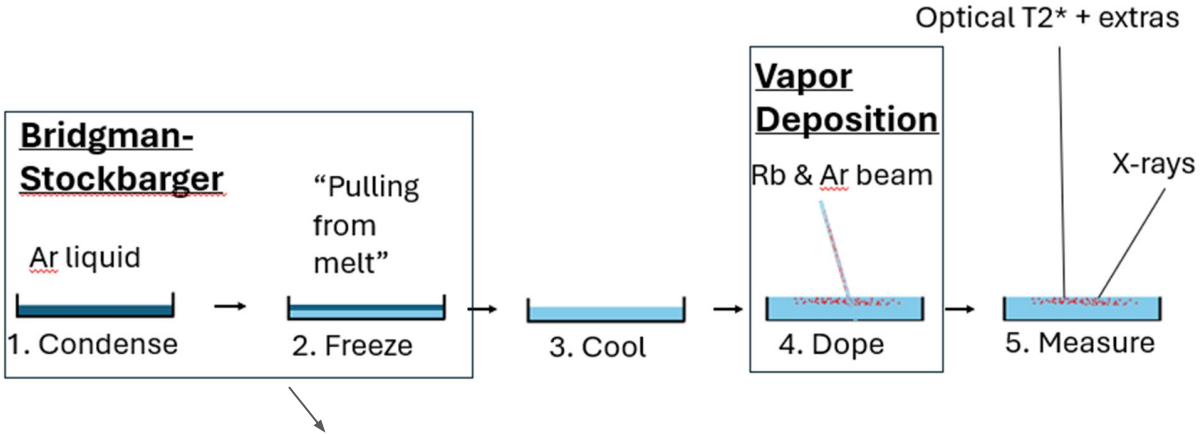




Ar Crystals!

Our Growth Method

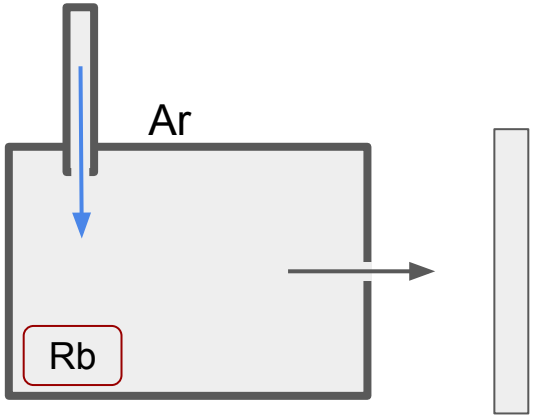




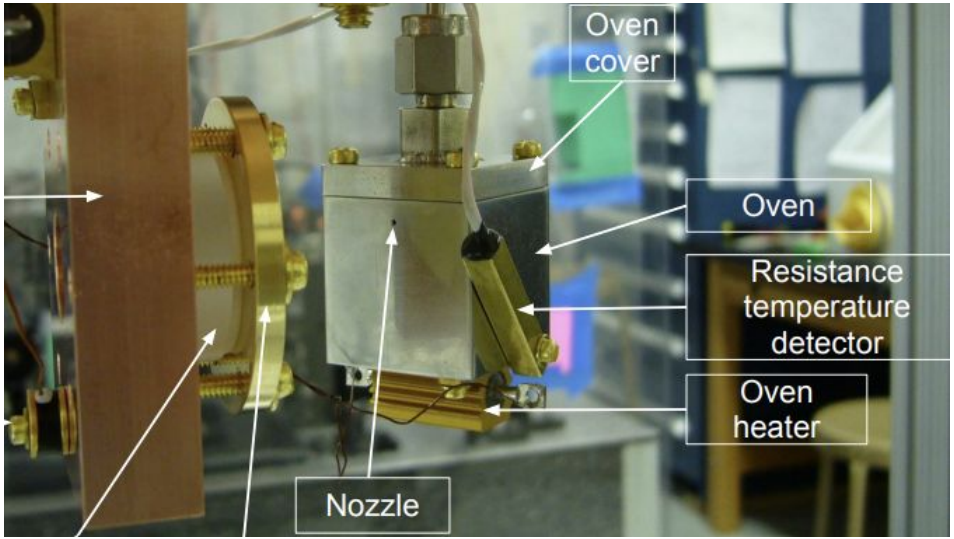


Ar Crystals!

Beam Source Constraints

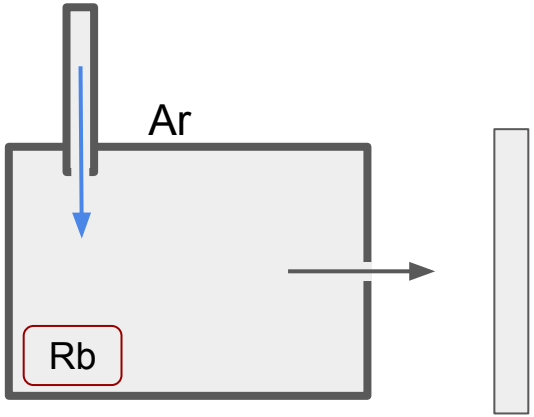


- Dopant Density $\propto P_{Rb}/P_{Ar}$
- Growth Rate $\propto P_{Rb} \cdot C$

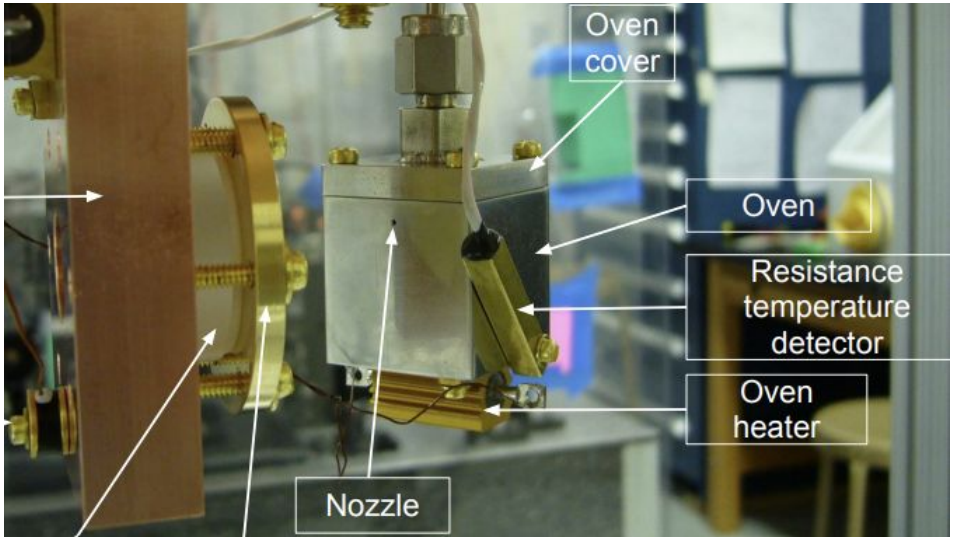


Regmi, 2013, M.S. thesis, University of Nevada, Reno.
Weinstein Group

Beam Source Constraints

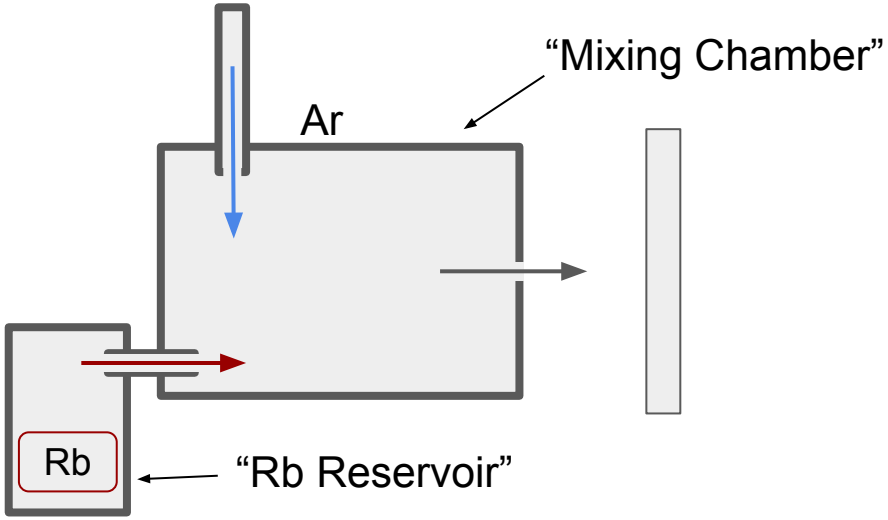


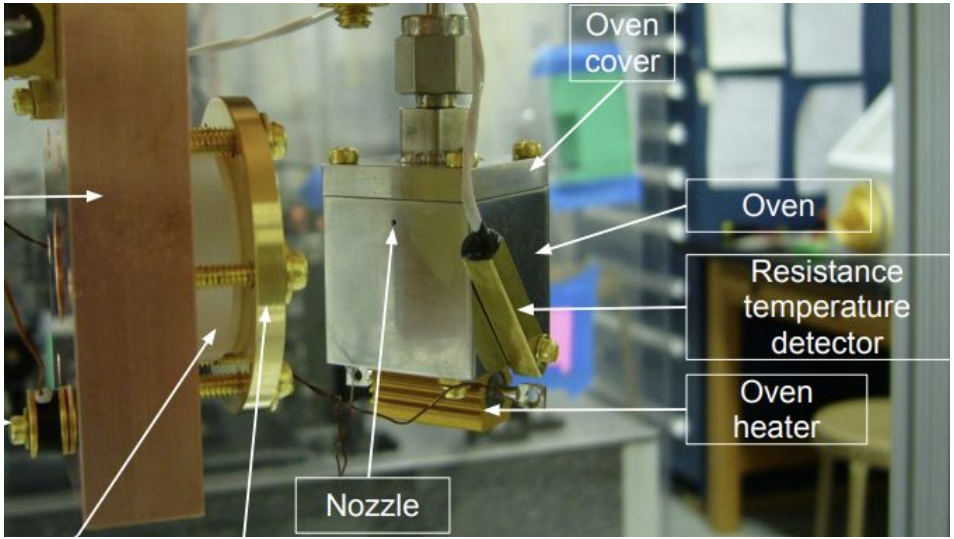
- Dopant Density $\propto P_{Rb}/P_{Ar}$
- Growth Rate $\propto P_{Rb} \cdot C$



Regmi, 2013, M.S. thesis, University of Nevada, Reno.
Weinstein Group

Beam Source Constraints

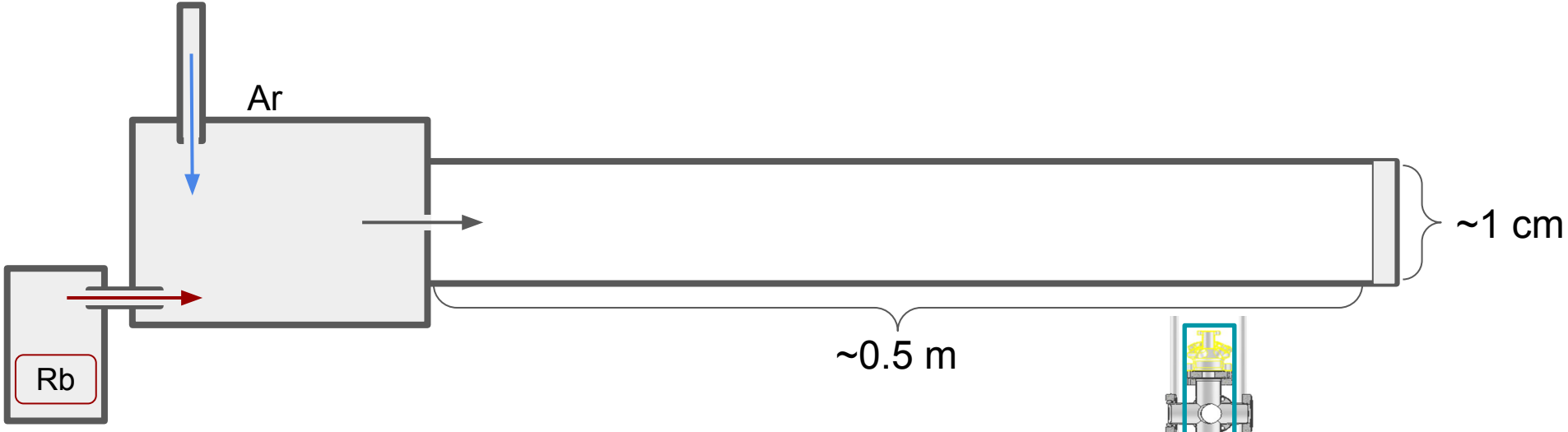




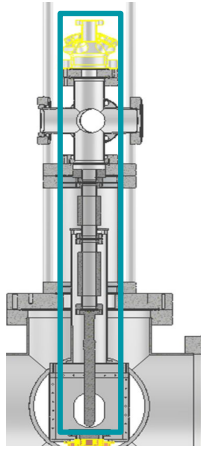
Regmi, 2013, M.S. thesis, University of Nevada, Reno.
Weinstein Group

- $P_{RbMc} \propto P_{RbRs} \frac{C_V}{C_V + C_N}$

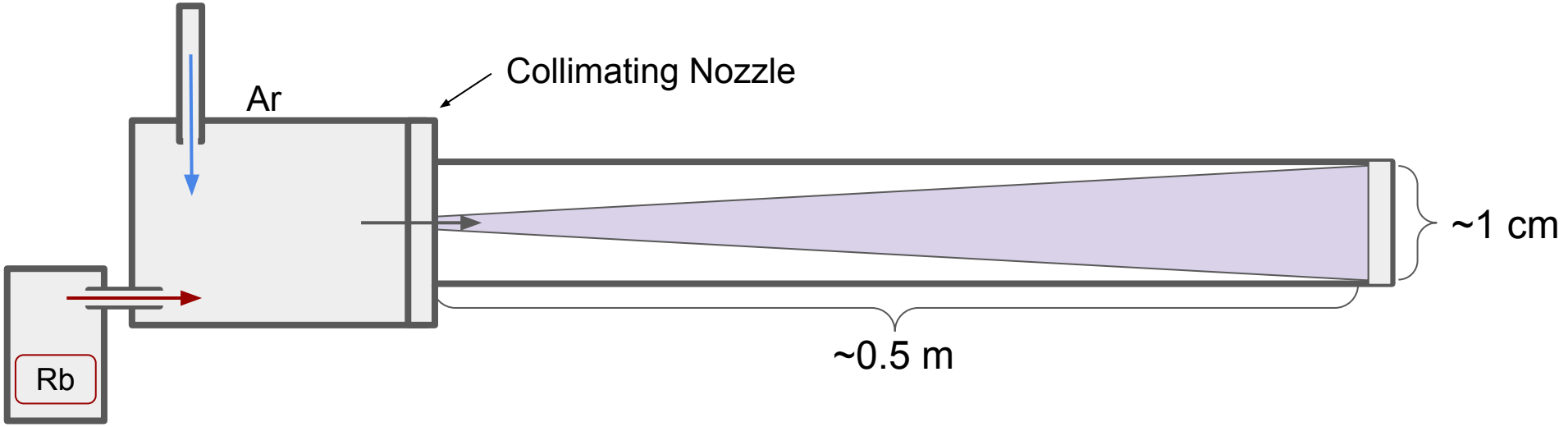
Beam Source Constraints



- Bridgman-Stockbarger ⇒ inner + outer chamber
- X-ray compatibility ⇒ strict spatial contraints
- ⇒ long path to reduce heat loads

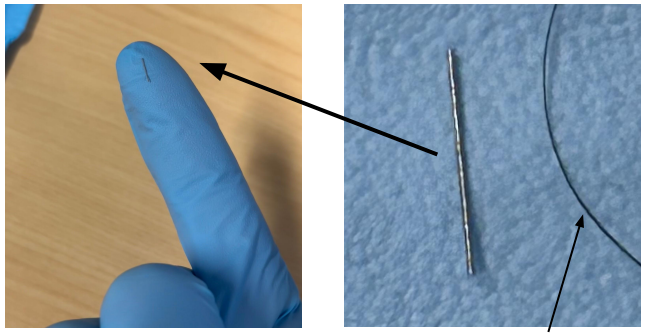


Beam Source Constraints



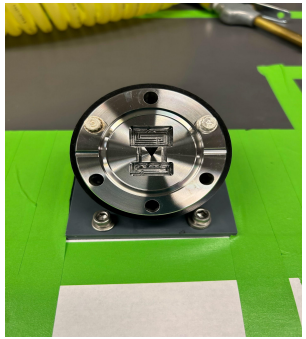
- Need large Ar flux
- Narrow divergence

Microcapillary Nozzle Array

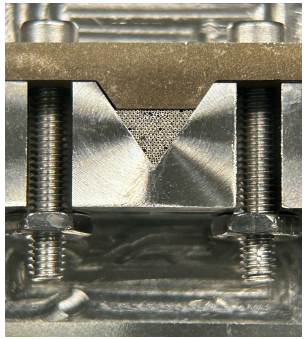


hair for reference

- 231, 100 μm ID, ~0.8 cm length microcapillaries
- Wire EDM cut 5ft microcapillary from MicroGroup
- Wire EDM cut 2.75” CF flange



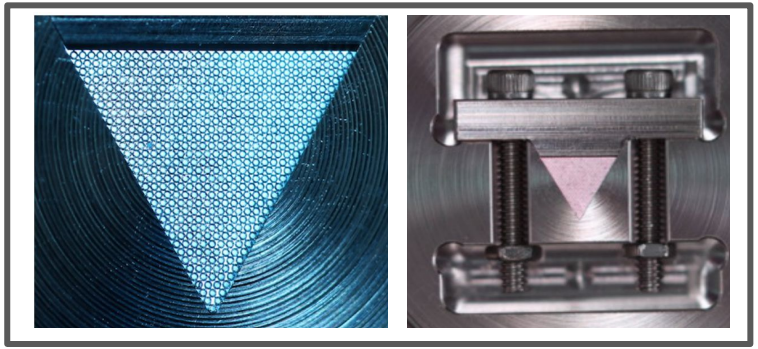
Assembly Setup



Oven Side



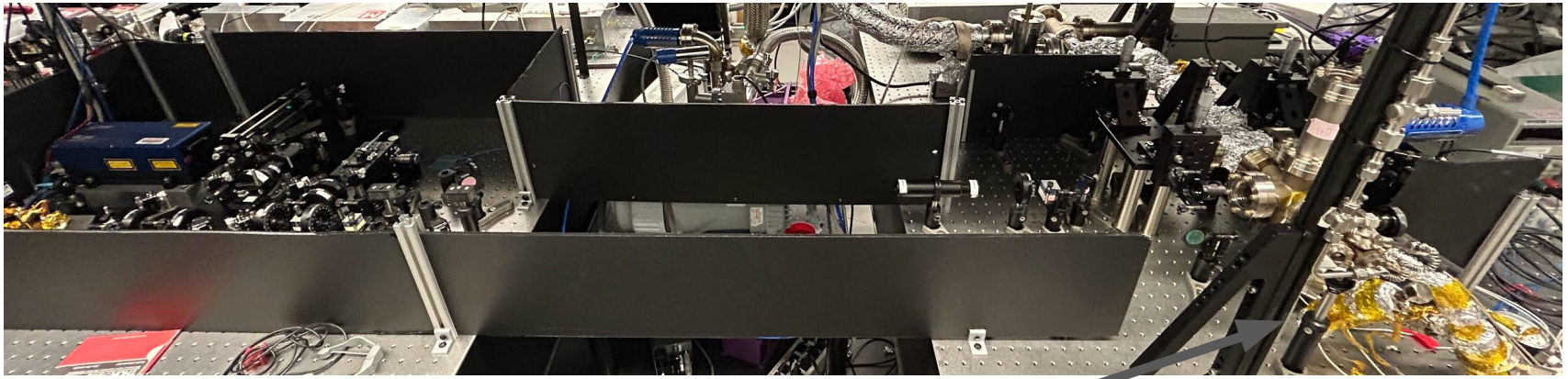
Growth Chamber Side

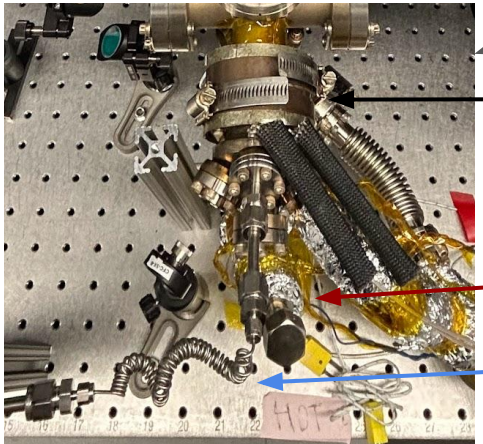


Original Design:

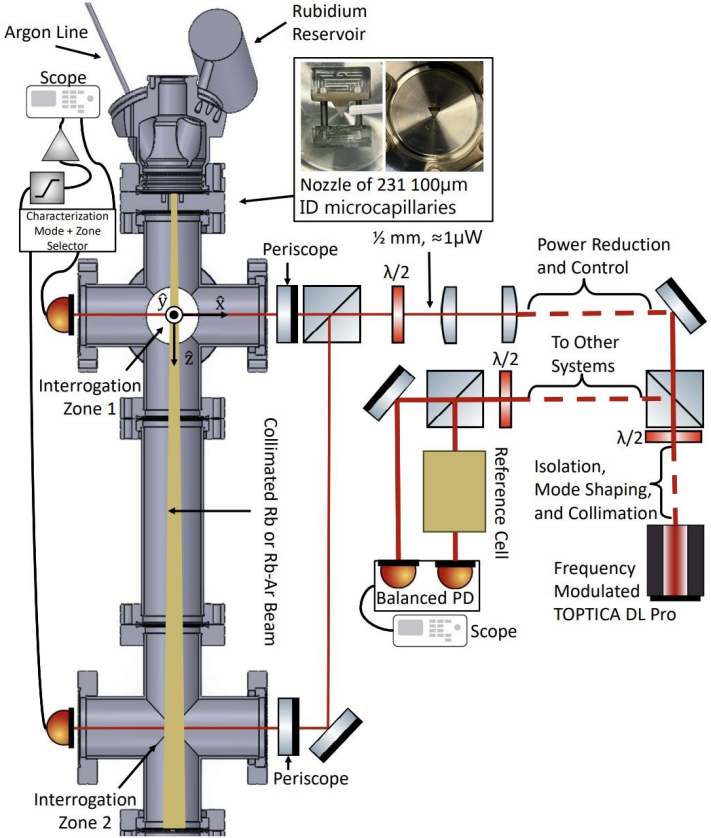
R. Senaratne et al., *Rev. Sci. Instrum.* **86**, 023105 (2015).

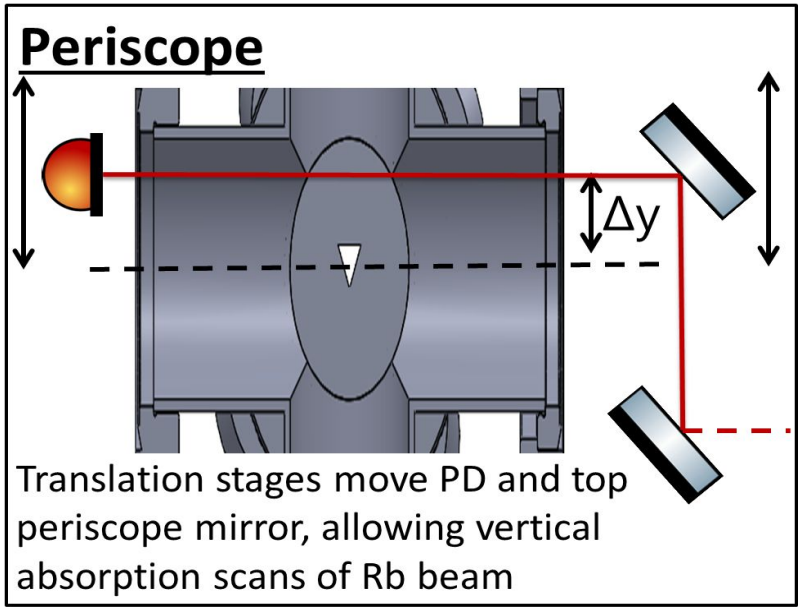
Characterizing Beam Source





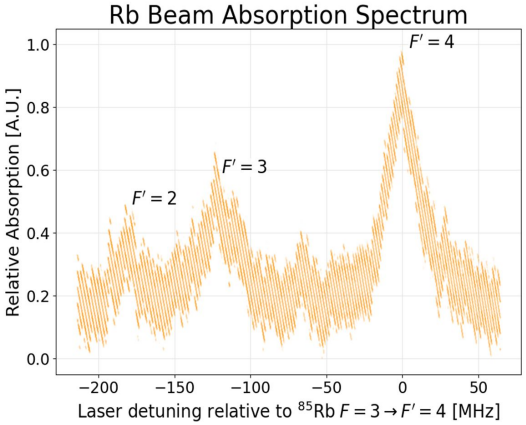
Characterizing Beam Source

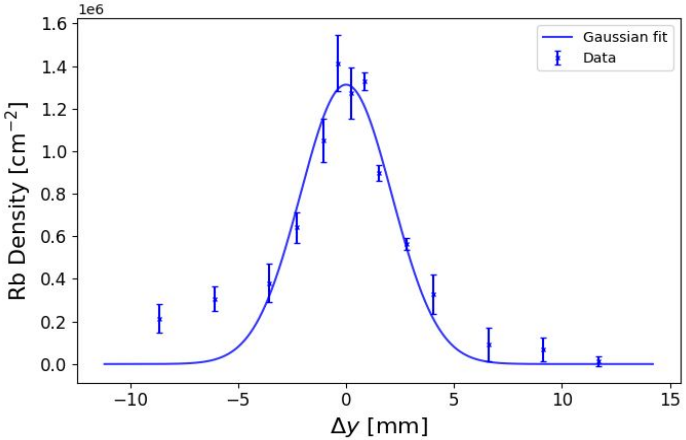




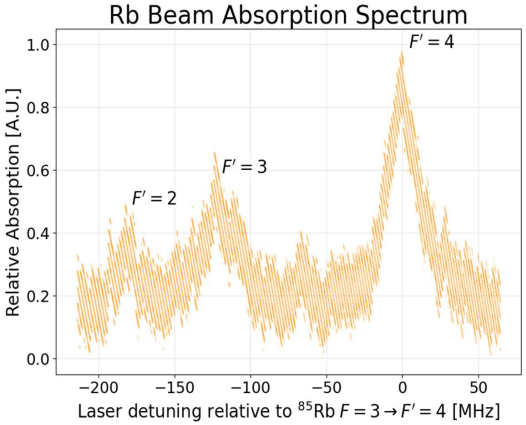
- $\lambda_{Rb} \leq \lambda_{Ar}$

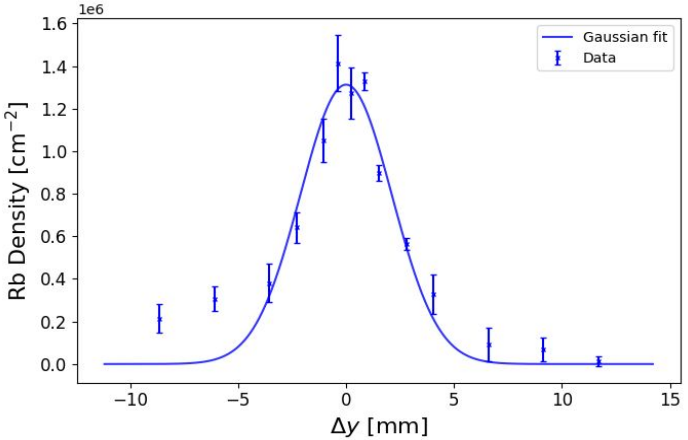
Characterizing Beam Source





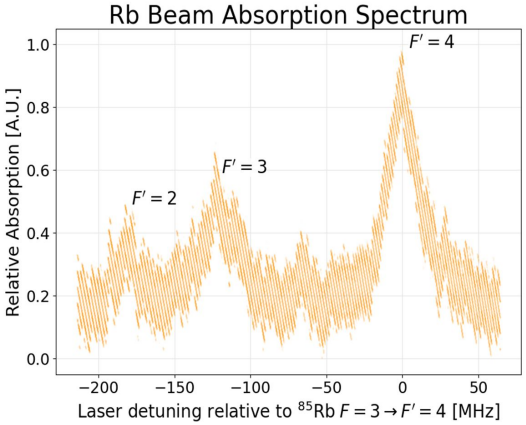
Characterizing Beam Source

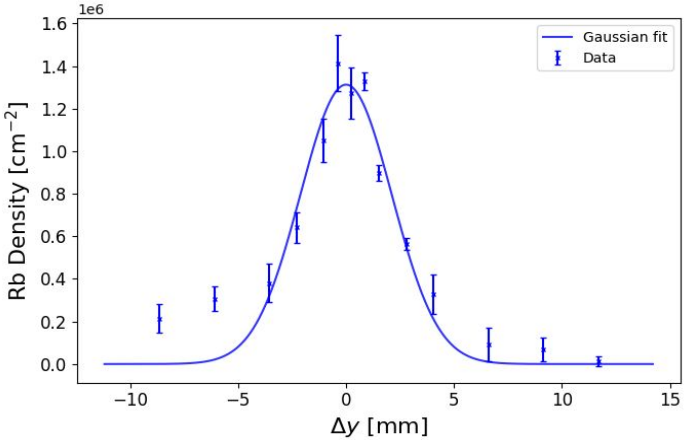




- $(\alpha)_{1/e} = 1 \pm 0.1$ degrees
- Rb Flux: $2.5 * 10^{10}$ atoms/s
- Ar Flux: $5 * 10^{15}$ atoms/s
- Dopant Density: 10^{17} dopants/cm^3

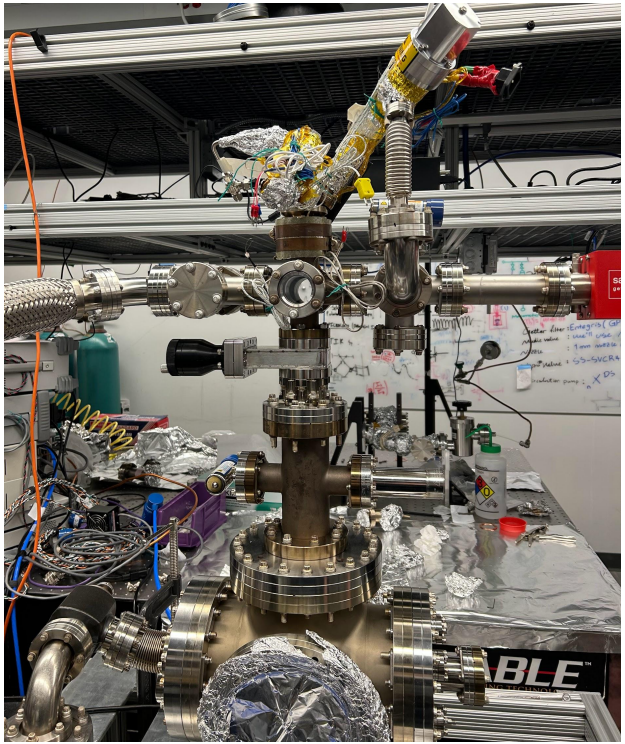
Characterizing Beam Source





- $(\alpha)_{1/e} = 1 \pm 0.1$ degrees
- Rb Flux: $2.5 * 10^{10}$ atoms/s
- Ar Flux: $5 * 10^{15}$ atoms/s
- Dopant Density: 10^{17} dopants/cm^3

Next Steps



- Grow Doped Crystals!!
- Measure crystal properties to check beam characterization
  - Growth Rate/Dopant Density can be directly measured
- Explore larger parameter space

Acknowledgements

The Odom Group


Brian Odom


Daniel Lutes


Joel Adhiambo


Haley Randall


Brandon Lofton

Kovachy Lab


Tim Kovachy


Sharika Saraf

ACME


Xing Fan

Argonne NATIONAL LABORATORY

NORTHWESTERN UNIVERSITY QUAECUMQUE SUNT VERA 1851

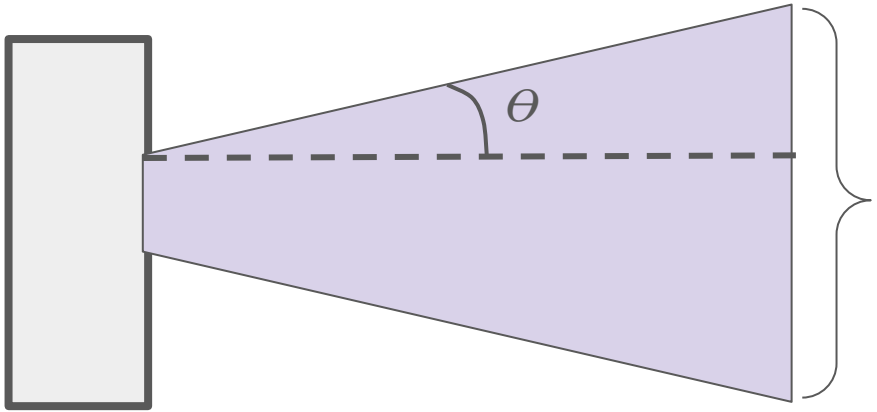
NSF

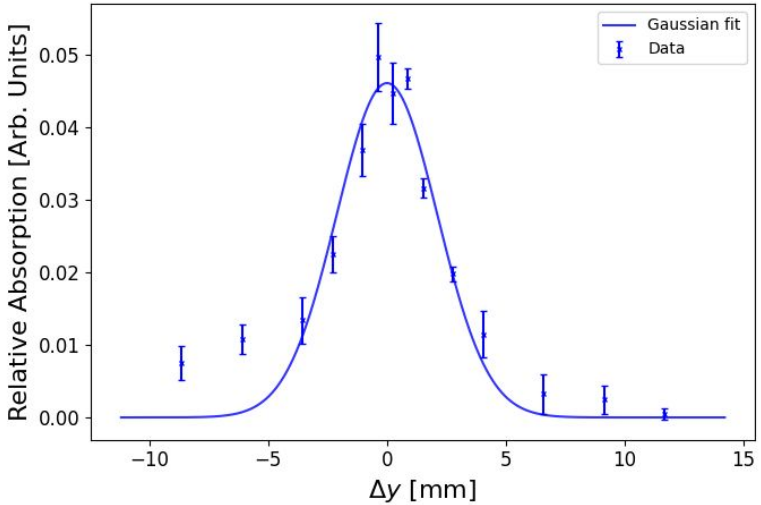


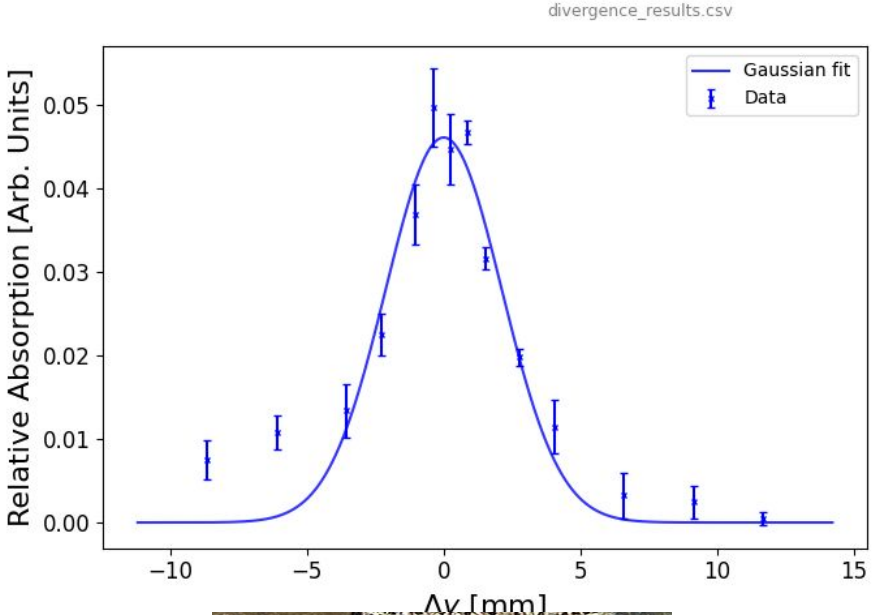
The study resulting in this presentation was assisted by a grant from the Undergraduate Research Grant Program which is administered by Northwestern University's Office of Undergraduate Research. However, the conclusions, opinions, and other statements in this presentation are the author's and not necessarily those of the sponsoring institution.

Supplemental Slides:

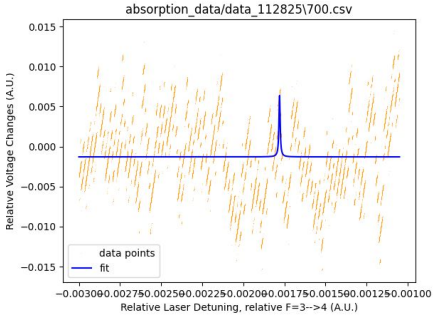
Divergence


θ
1/e width


Gaussian fit
Data
Relative Absorption [Arb. Units]
0.00
0.01
0.02
0.03
0.04
0.05
−10
−5
0
5
10
15
Δy [mm]

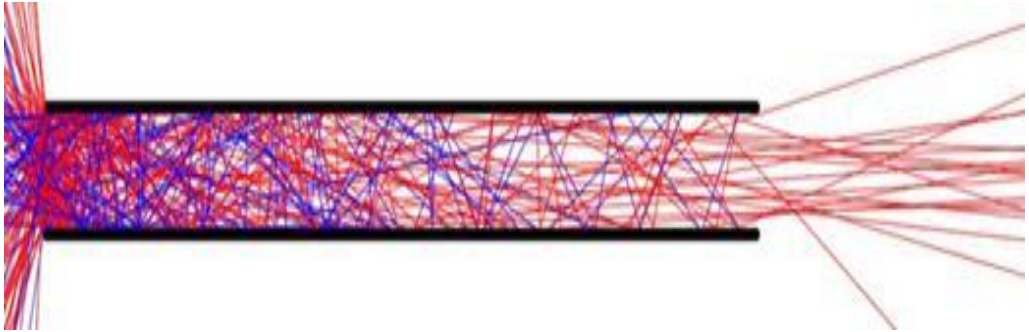

divergence_results.csv
Gaussian fit
Data
Relative Absorption [Arb. Units]
0.05
0.04
0.03
0.02
0.01
0.00
−10
−5
0
5
10
15
Δy [mm]




absorption_data/data_112825\700.csv
Relative Voltage Changes (A.U.)
0.015
0.010
0.005
0.000
−0.005
−0.010
−0.015
data points
fit
−0.00300
−0.00275
−0.00250
−0.00225
−0.00200
−0.00175
−0.00150
−0.00125
−0.00100
Relative Laser Detuning, relative F=3-->4 (A.U.)

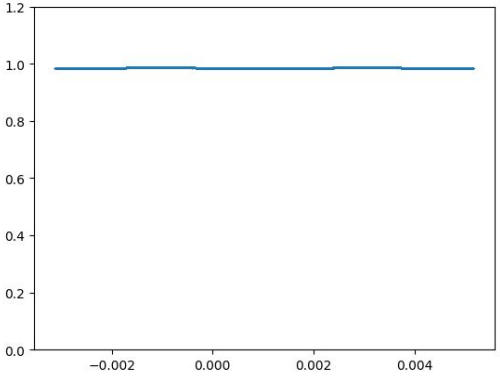
Rb-Ar

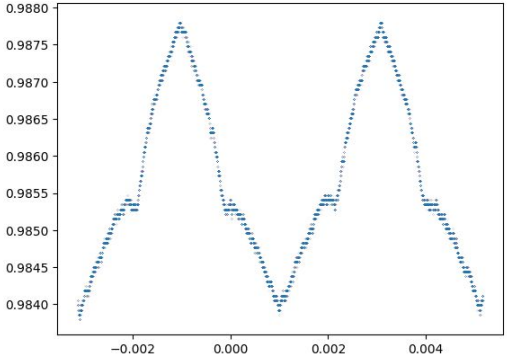
Smth about Ar and Rb being same divergence….

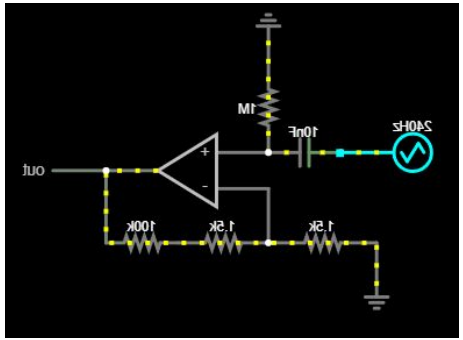


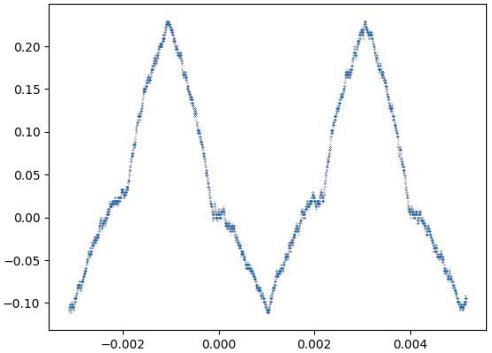
- λ = k_B T/(sqrt[2]pi*d^2*P)
- Prb << Par
- λrb/λar ~ d_ar^2/d_rb^2 ~½

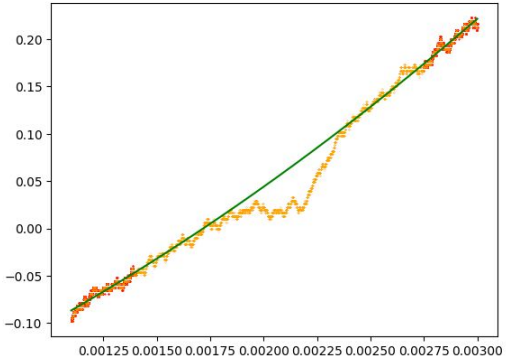
- Want λ ~> L > d to get aspect ratio like divergence.
- -> λrb will exit ideal regime of divergence set by aspect ratio before λar does

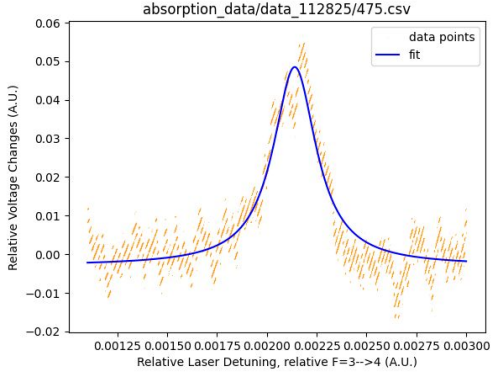

1.2
1.0
0.8
0.6
0.4
0.2
0.0
−0.002
0.000
0.002
0.004

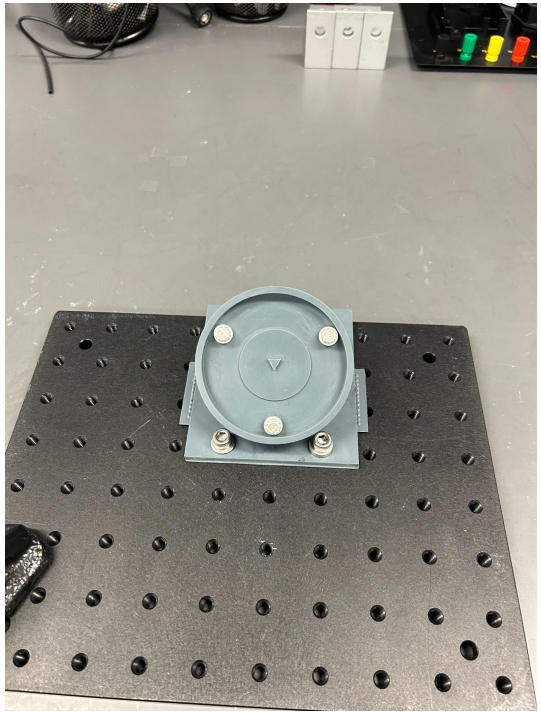

0.9880
0.9875
0.9870
0.9865
0.9860
0.9855
0.9850
0.9845
0.9840
−0.002
0.000
0.002
0.004

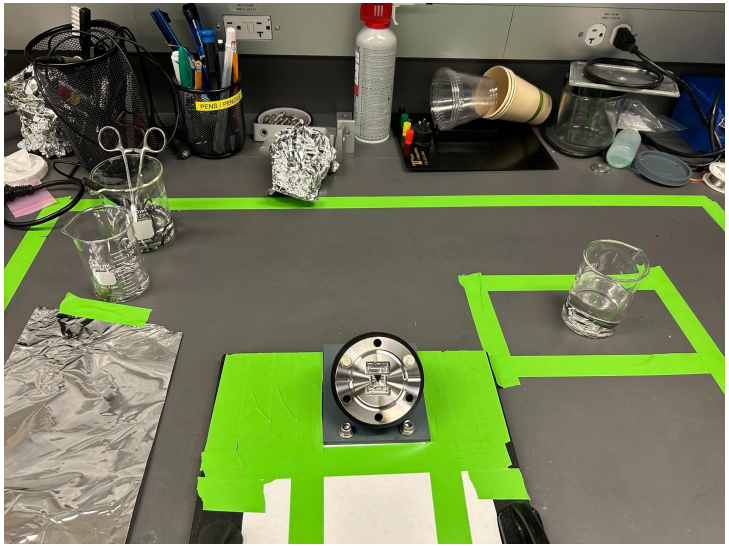

1M
10nF
240Hz
out
+
-
100k
1.5k
1.5k

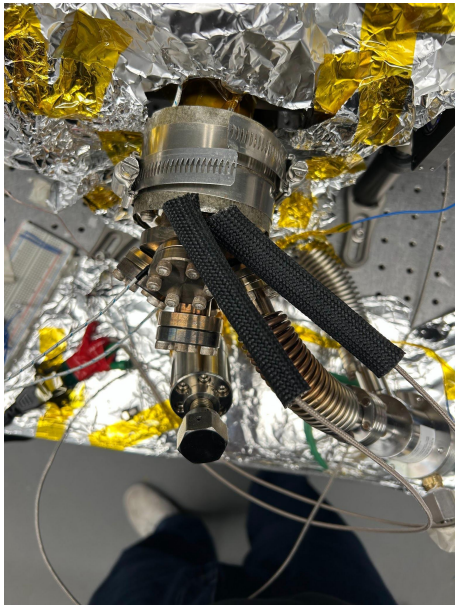

0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.002
0.000
0.002
0.004


0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
0.00125 0.00150 0.00175 0.00200 0.00225 0.00250 0.00275 0.00300


absorption_data/data_112825/475.csv
data points
fit
Relative Voltage Changes (A.U.)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
0.00125 0.00150 0.00175 0.00200 0.00225 0.00250 0.00275 0.00300
Relative Laser Detuning, relative F=3-->4 (A.U.)







Current Doped Cryogenic Crystals

Cs in He:

$T2^* = 0.1$ s

10^9 atoms/cm^3

Kanorsky et al.,
Phys. Rev. A **54**, R1010 (1996)

↓

Demonstrates one can achieve long coherence times in a cryogenic crystal

Rb in Ne:

$T2 = 0.1$ s

$T2^* = 10$ μs

10^{16} atoms/cm^3

J.D. Weinstein et al.,
Phys. Rev. A **104**, 032611 (2021)

→ Not enough dopants due to high helium pressure required for growth

→ Many dopants

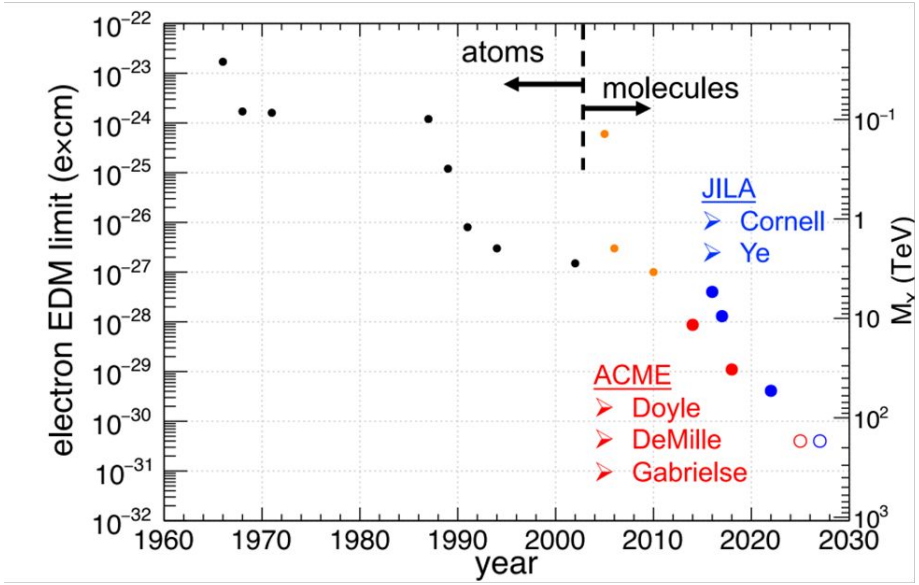
↓

Large discrepancy between T2 and T2* due to inhomogeneous broadening, thought to be from grain boundaries

↓

Will address this by growing single crystal seeds and then performing epitaxy

Progress in EDM Sensitivity



Science **343**, 269 (2014) Phys. Rev. Lett. **119** 153001
Nature **562**, 355 (2018) Science, **381**,46 (2023)

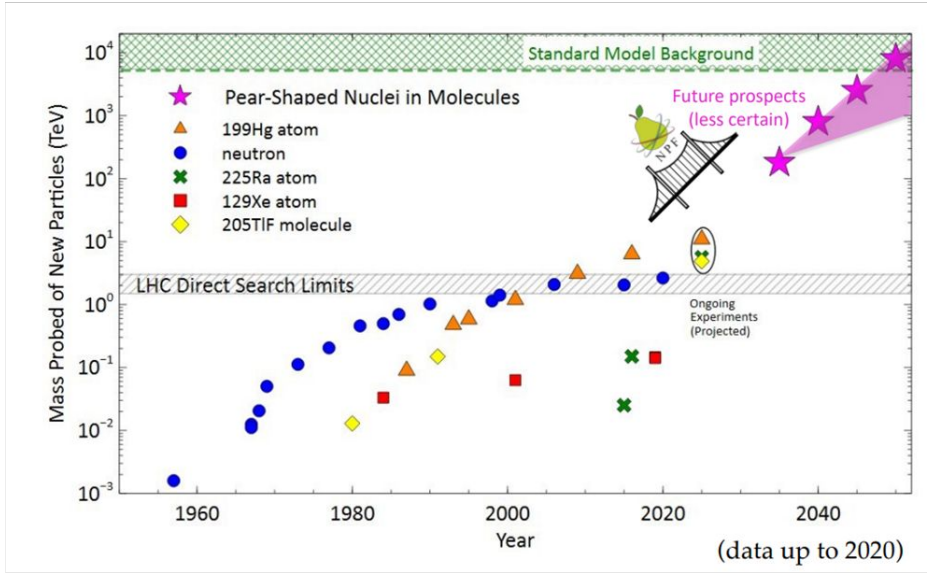
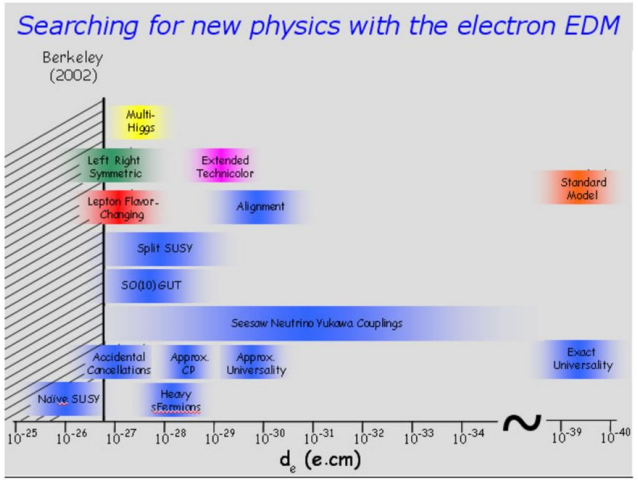
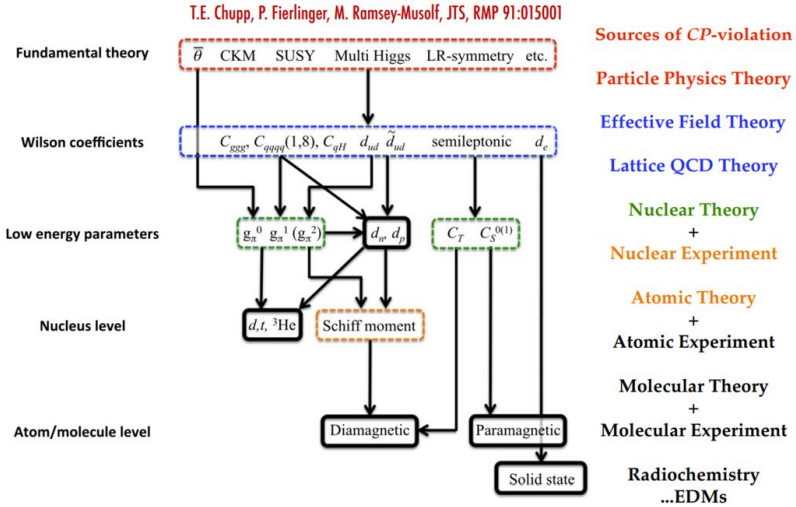
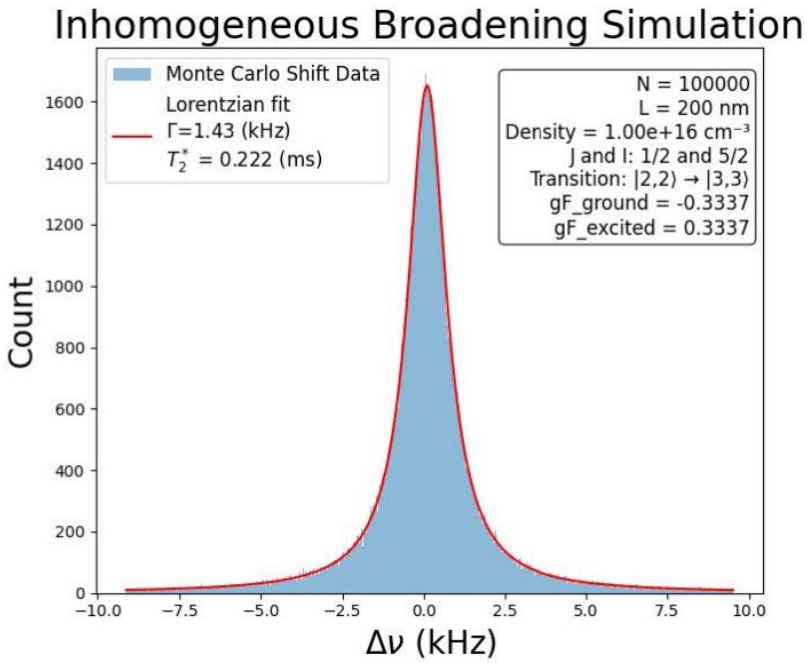


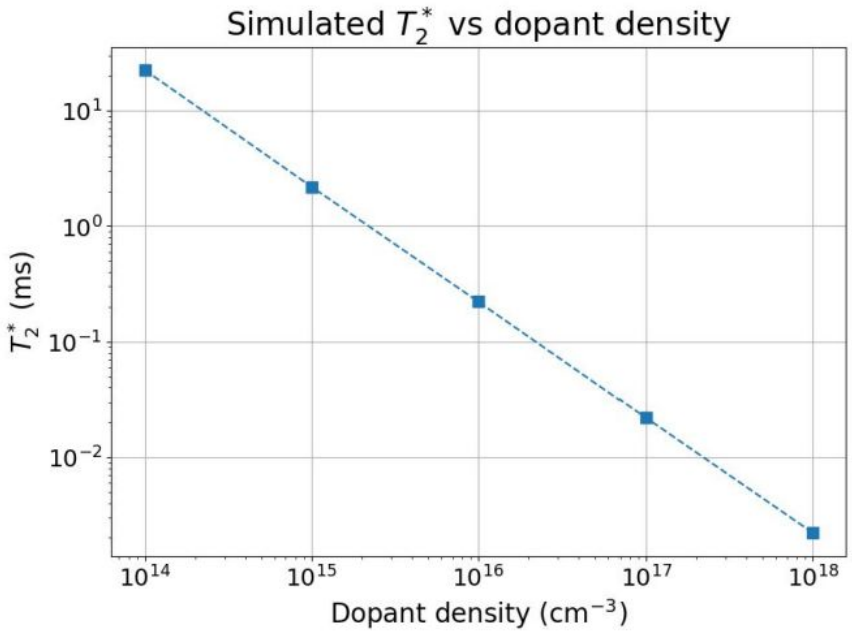
Figure from Jaideep Taggart Singh 2024 NNSS



Predicted electron electric dipole moments (compiled by D. DeMille)




Inhomogeneous Broadening Simulation
Monte Carlo Shift Data
Lorentzian fit
Γ=1.43 (kHz)
T_2^* = 0.222 (ms)
N = 100000
L = 200 nm
Density = 1.00e+16 cm⁻³
J and I: 1/2 and 5/2
Transition: |2,2⟩ → |3,3⟩
gF_ground = -0.3337
gF_excited = 0.3337
Count
Δν (kHz)


Simulated T_2^* vs dopant density
T_2^* (ms)
Dopant density (cm^{-3})